

spin conversing excitation TDA analytic gradient

王宏博

2026.6.14

1 excited state analytic gradient basis

激发态解析梯度可以写为基态能量的梯度加激发能的梯度，即

$$\frac{dE_{exc}}{d\xi} = \frac{dE_g}{d\xi} + \frac{d\omega}{d\xi} \quad (1.1)$$

基态能量的梯度在现有程序中容易计算，这里暂时不展开，以后补充这块内容。要计算激发能的梯度，首先要得到激发能，对于 UTDDFT，先求解 Casida equation

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B} & \mathbf{A} \end{pmatrix} \begin{pmatrix} \mathbf{X} \\ \mathbf{Y} \end{pmatrix} = \omega \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \mathbf{X} \\ \mathbf{Y} \end{pmatrix} \quad (1.2)$$

得到 ω

$$\omega = \frac{1}{2} [\mathbf{R}^T (\mathbf{A} + \mathbf{B}) \mathbf{R} + \mathbf{L}^T (\mathbf{A} - \mathbf{B}) \mathbf{L}] \quad (1.3)$$

$$\mathbf{R} = (\mathbf{X} + \mathbf{Y}) \quad (1.4)$$

$$\mathbf{L} = (\mathbf{X} - \mathbf{Y}) \quad (1.5)$$

$$A_{ia,jb}^{\sigma\tau} = \delta_{\sigma\tau} (\delta_{ij} F_{ab}^{\sigma} - F_{ij}^{\sigma} \delta_{ab}) + K_{ia,jb}^{\sigma,\tau} \quad (1.6)$$

$$B_{ia,jb}^{\sigma\tau} = K_{ia,bj}^{\sigma\tau} \quad (1.7)$$

$$K_{pq,rs}^{\sigma\tau} = (p^{\sigma} q^{\sigma} | s^{\tau} r^{\tau}) - c_x \delta_{\sigma\tau} (p^{\sigma} r^{\tau} | s^{\sigma} q^{\tau}) + f_{pq,sr}^{xc,\sigma\tau} [\rho] \quad (1.8)$$

$$f_{pq,sr}^{xc,\sigma\tau} [\rho] = \frac{\delta^2 E_{xc}}{\delta \rho_{pq}^{\sigma} \delta \rho_{sr}^{\tau}} \quad (1.9)$$

对于 UTDA，Casida equation 退化为

$$\mathbf{A} \mathbf{X} = \omega \mathbf{X} \quad (1.10)$$

$$\omega = \mathbf{X}^T \mathbf{A} \mathbf{X} \quad (1.11)$$

$\mathbf{A}$ 与 UTDDFT 相同。

对于 ROKS 参考态下，UTDA 的 Casida equation 与 UKS 参考态下相同， $\mathbf{A}$ 也与 UKS 参考态下相同，注意，Fock 与 UTDA 不同，后面 Lagrange Multiplier method 中的约束也与 UTDA 不同。

$$\mathbf{A} \mathbf{X} = \omega \mathbf{X} \quad (1.12)$$

$$\omega = \mathbf{X}^T \mathbf{A} \mathbf{X} \quad (1.13)$$

2 UKS with U-TDDFT analytic gradient

首先给出 UKS 参考态下 spin conversing excitation 的解析梯度。使用 Lagrange multiplier method，有三个约束需要满足：

1. 正交

$$\mathbf{X}^T \mathbf{X} - \mathbf{Y}^T \mathbf{Y} = 1 \quad (2.1)$$

2. Brillouin 定理

$$F_{ia}^\sigma = F_{ai}^\sigma = 0 \quad (2.2)$$

3. 正则轨道

$$S_{pq}^\sigma = \delta_{pq} \quad (2.3)$$

所以拉格朗日量可以写为

$$\mathcal{L}[\mathbf{X}, \mathbf{Y}, \mathbf{C}, \Omega, \mathbf{Z}, \mathbf{W}] = \omega - \Omega(\mathbf{X}^T \mathbf{X} - \mathbf{Y}^T \mathbf{Y} - 1) + \sum_{ai\sigma} Z_{ai}^\sigma F_{ai}^\sigma - \sum_{pq\sigma} W_{pq}^\sigma (S_{pq}^\sigma - \delta_{pq}) \quad (2.4)$$

其中 Ω , Z_{ai}^σ , W_{pq}^σ 都是拉格朗日乘子，拉格朗日乘子法要求

$$\frac{\partial \mathcal{L}}{\partial X_{ia}^\sigma} = 0, \quad \frac{\partial \mathcal{L}}{\partial Y_{ia}^\sigma} = 0, \quad \frac{\partial \mathcal{L}}{\partial C_{\mu p}^\sigma} = 0, \quad \frac{\partial \mathcal{L}}{\partial \Omega} = 0, \quad \frac{\partial \mathcal{L}}{\partial Z_{ai}^\sigma} = 0, \quad \frac{\partial \mathcal{L}}{\partial W_{pq}^\sigma} = 0 \quad (2.5)$$

所以

$$\frac{d\omega}{d\xi} = \frac{\partial \mathcal{L}}{\partial \xi} + \frac{\partial \mathcal{L}}{\partial X_{ia}^\sigma} \frac{\partial X_{ia}^\sigma}{\partial \xi} + \frac{\partial \mathcal{L}}{\partial Y_{ia}^\sigma} \frac{\partial Y_{ia}^\sigma}{\partial \xi} + \frac{\partial \mathcal{L}}{\partial C_{\mu p}^\sigma} \frac{\partial C_{\mu p}^\sigma}{\partial \xi} + \frac{\partial \mathcal{L}}{\partial \Omega} \frac{\partial \Omega}{\partial \xi} + \frac{\partial \mathcal{L}}{\partial Z_{ai}^\sigma} \frac{\partial Z_{ai}^\sigma}{\partial \xi} + \frac{\partial \mathcal{L}}{\partial W_{pq}^\sigma} \frac{\partial W_{pq}^\sigma}{\partial \xi} = \frac{d\mathcal{L}}{d\xi} \quad (2.6)$$

即激发能的梯度等于拉格朗日量对坐标的偏导数。将这个偏导数展开

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \xi} = \sum_{\mu\nu\sigma} \left[(T_{\mu\nu}^\sigma + Z_{\mu\nu}^{S,\sigma}) \frac{\partial h_{\mu\nu}^{core}}{\partial \xi} + (T_{\mu\nu}^\sigma + Z_{\mu\nu}^{S,\sigma}) \frac{\partial g_{\mu\nu}^\sigma[\mathbf{D}]}{\partial \xi} + (T_{\mu\nu}^\sigma + Z_{\mu\nu}^{S,\sigma}) v_{\mu\nu}^{xc,\sigma,(\xi)}[\rho] \right. \\ \left. + R_{\mu\nu}^{S,\sigma} \frac{\partial g_{\mu\nu}^\sigma[\mathbf{R}^S]}{\partial \xi} + L_{\mu\nu}^{A,\sigma} \frac{\partial g_{\mu\nu}^\sigma[\mathbf{L}^A]}{\partial \xi} + \sum_{\kappa\lambda\tau} R_{\mu\nu}^\sigma R_{\kappa\lambda}^\tau f_{\mu\nu,\kappa\lambda}^{xc,\sigma\tau,(\xi)}[\rho] - \frac{\partial S_{\mu\nu}^\sigma}{\partial \xi} W_{\mu\nu}^\sigma \right] \end{aligned} \quad (2.7)$$

其中

$$\begin{aligned} T_{\mu\nu}^\sigma &= \sum_{ab} C_{\mu a}^\sigma T_{ab}^\sigma C_{\nu b}^\sigma + \sum_{ij} C_{\mu i}^\sigma T_{ij}^\sigma C_{\nu j}^\sigma \\ T_{ab}^\sigma &= \frac{1}{2} \sum_i (R_{ia}^\sigma R_{ib}^\sigma + L_{ia}^\sigma L_{ib}^\sigma) \\ T_{ij}^\sigma &= -\frac{1}{2} \sum_i (R_{ia}^\sigma R_{ja}^\sigma + L_{ia}^\sigma L_{ja}^\sigma) \\ Z_{\mu\nu}^{S,\sigma} &= \frac{1}{2} (Z_{\mu\nu}^\sigma + Z_{\nu\mu}^\sigma) \\ R_{\mu\nu}^{S,\sigma} &= \frac{1}{2} (R_{\mu\nu}^\sigma + R_{\nu\mu}^\sigma) \\ L_{\mu\nu}^{A,\sigma} &= \frac{1}{2} (L_{\mu\nu}^\sigma - L_{\nu\mu}^\sigma) \\ g_{\mu\nu}^\sigma[\mathbf{V}] &= \sum_{\kappa\lambda\tau} [(\mu\nu|\kappa\lambda) - c_x \delta_{\sigma\tau} (\mu\lambda|\kappa\nu)] V_{\kappa\lambda} \\ v_{\mu\nu}^{xc,\sigma,(\xi)}[\rho] &= \frac{\partial v_{\mu\nu}^{xc,\sigma}[\rho]}{\partial \xi} \\ f_{\mu\nu,\kappa\lambda}^{xc,\sigma\tau,(\xi)}[\rho] &= \frac{\partial f_{\mu\nu,\kappa\lambda}^{xc,\sigma\tau}[\rho]}{\partial \xi} \end{aligned} \quad (2.8)$$

为了后续公式的简洁，定义

$$G_{\mu\nu}^\sigma[\mathbf{V}] = g_{\mu\nu}^\sigma[\mathbf{V}] + f_{\mu\nu}^{xc,\sigma}[\mathbf{V}] \quad (2.9)$$

接下来需要通过 $\partial\mathcal{L}/\partial C_{\mu p}^\sigma$ 来计算 $Z_{\mu\nu}^{S,\sigma}$ 和 $W_{\mu\nu}^\sigma$ ，用下面的形式计算能够方便的得到这两个量

$$\sum_{\mu'} \frac{\partial\mathcal{L}}{\partial C_{\mu'p'}^{\sigma'}} C_{\mu'q'}^{\sigma'} = 0 \quad (2.10)$$

为了避免公式过长，首先计算

$$Q_{p'q'}^{\sigma'} = \sum_{\mu'} \frac{\partial\omega}{\partial C_{\mu'p'}^{\sigma'}} C_{\mu'q'}^{\sigma'} \quad (2.11)$$

$$\begin{aligned} Q_{i'j'}^{\sigma'} &= 2 \left(\sum_j T_{i'j}^{\sigma'} F_{jj'}^{\sigma'} + G_{i'j'}[\mathbf{T}] + \sum_a R_{i'a}^{\sigma'} G_{ja}^{\sigma'}[\mathbf{R}^S] + \sum_a L_{i'a}^{\sigma'} g_{ja}^{\sigma'}[\mathbf{L}^A] + g_{i'j'}^{xc,\sigma'}[\mathbf{R}, \mathbf{R}] \right) \\ Q_{a'b'}^{\sigma'} &= 2 \left(\sum_b T_{a'b}^{\sigma'} F_{bb'}^{\sigma'} + \sum_i R_{ia'}^{\sigma'} G_{ib'}^{\sigma'}[\mathbf{R}^S] + \sum_i L_{ia'}^{\sigma'} g_{ib'}^{\sigma'}[\mathbf{L}^A] \right) \\ Q_{i'a'}^{\sigma'} &= 2 \left(\sum_j T_{i'j}^{\sigma'} F_{a'j}^{\sigma'} + G_{i'a'}[\mathbf{T}] + \sum_a R_{i'a}^{\sigma'} G_{a'a}^{\sigma'}[\mathbf{R}^S] + \sum_a L_{i'a}^{\sigma'} g_{a'a}^{\sigma'}[\mathbf{L}^A] + g_{i'a'}^{xc,\sigma'}[\mathbf{R}, \mathbf{R}] \right) \\ Q_{a'i'}^{\sigma'} &= 2 \left(\sum_b T_{a'b}^{\sigma'} F_{i'b}^{\sigma'} + \sum_i R_{ia'}^{\sigma'} G_{ii'}^{\sigma'}[\mathbf{R}^S] + \sum_i L_{ia'}^{\sigma'} g_{ii'}^{\sigma'}[\mathbf{L}^A] \right) \end{aligned} \quad (2.12)$$

接下来给出完整的 $\sum_{\mu'} \frac{\partial\mathcal{L}}{\partial C_{\mu'p'}^{\sigma'}} C_{\mu'q'}^{\sigma'}$

$$\begin{aligned} \sum_{\mu'} \frac{\partial\mathcal{L}}{\partial C_{\mu'i'}^{\sigma'}} C_{\mu'j'}^{\sigma'} &= Q_{i'j'}^{\sigma'} + 2G_{i'j'}^{\sigma'}[\mathbf{Z}^S] - 2W_{i'j'}^{\sigma'} \\ \sum_{\mu'} \frac{\partial\mathcal{L}}{\partial C_{\mu'a'}^{\sigma'}} C_{\mu'b'}^{\sigma'} &= Q_{a'b'}^{\sigma'} - 2W_{a'b'}^{\sigma'} \\ \sum_{\mu'} \frac{\partial\mathcal{L}}{\partial C_{\mu'i'}^{\sigma'}} C_{\mu'a'}^{\sigma'} &= Q_{i'a'}^{\sigma'} + \sum_a Z_{ai'}^{\sigma'} F_{a'a}^{\sigma'} + 2G_{i'a'}^{\sigma'}[\mathbf{Z}^S] - 2W_{i'a'}^{\sigma'} \\ \sum_{\mu'} \frac{\partial\mathcal{L}}{\partial C_{\mu'a'}^{\sigma'}} C_{\mu'i'}^{\sigma'} &= Q_{a'i'}^{\sigma'} + \sum_i Z_{a'i}^{\sigma'} F_{i'i}^{\sigma'} - 2W_{a'i'}^{\sigma'} \end{aligned} \quad (2.13)$$

由于拉格朗日乘子法要求 $\frac{\partial\mathcal{L}}{\partial C_{\mu'p'}^{\sigma'}} = 0$ ，并且由于 $S_{pq}^\sigma = S_{qp}^\sigma$ ，有 $W_{pq}^\sigma = W_{qp}^\sigma \rightarrow W_{i'a'}^{\sigma'} = W_{a'i'}^{\sigma'}$ ，进而可以得到 Z vector equation

$$-(Q_{i'a'}^{\sigma'} - Q_{a'i'}^{\sigma'}) = \sum_a Z_{ai'}^{\sigma'} F_{a'a}^{\sigma'} + 2G_{i'a'}^{\sigma'}[\mathbf{Z}^S] - \sum_i Z_{a'i}^{\sigma'} F_{i'i}^{\sigma'} \quad (2.14)$$

求解 Z vector equation 可以得到 Z_{ai}^σ ，进而可以得到 W_{pq}^σ

$$\begin{aligned} W_{i'j'}^{\sigma'} &= \frac{1}{2} Q_{i'j'}^{\sigma'} + G_{i'j'}^{\sigma'} \\ W_{a'b'}^{\sigma'} &= \frac{1}{2} Q_{a'b'}^{\sigma'} \\ W_{i'a'}^{\sigma'} &= \frac{1}{2} Q_{i'a'}^{\sigma'} + \frac{1}{2} \sum_a Z_{ai'}^{\sigma'} F_{a'a}^{\sigma'} + G_{i'a'}^{\sigma'} \\ W_{a'i'}^{\sigma'} &= \frac{1}{2} Q_{a'i'}^{\sigma'} + \frac{1}{2} \sum_i Z_{a'i}^{\sigma'} F_{i'i}^{\sigma'} \end{aligned} \quad (2.15)$$

有了 $W_{pq}^{\sigma'}$ 和 $Z_{pq}^{\sigma'}$ 带入 (2.7)，然后加基态能量的梯度即可完成计算。

3 ROKS with U-TDA analytic gradient

这一节推导 ROKS 参考态下 spin conversing 激发能的解析梯度，同样使用 Lagrangian Multiplier method，与 UKS 参考态下的 U-TDA 不同，三个约束条件变为

1. 正交（由于考虑 TDA，所以仅剩 $\mathbf{X}$ ）

$$\mathbf{X}^T \mathbf{X} = 1 \quad (3.1)$$

2. Brillouin 定理发生变化

$$F_{ai}^\alpha + F_{ai}^\beta = 0 \quad F_{at}^\alpha = 0 \quad F_{it}^\beta = 0 \quad (3.2)$$

3. 正则轨道

$$S_{pq} = \delta_{pq} \quad (3.3)$$

所以拉格朗日量可以写为

$$\mathcal{L}[\mathbf{X}, \mathbf{C}, \Omega, \mathbf{Z}, \mathbf{W}] = \omega - \Omega \left(\sum_{ia} X_{ia} X_{ia} - 1 \right) + \sum_{ai} Z_{ai} (F_{ai}^\alpha + F_{ai}^\beta) + \sum_{at} Z_{at} F_{at}^\alpha + \sum_{ti} Z_{ti} F_{ti}^\beta - \sum_{pq} W_{pq} (S_{pq} - \delta_{pq}) \quad (3.4)$$

同样拉格朗日乘子法要求

$$\frac{\partial \mathcal{L}}{\partial X_{ia}} = 0, \quad \frac{\partial \mathcal{L}}{\partial C_{\mu p}} = 0, \quad \frac{\partial \mathcal{L}}{\partial \Omega} = 0, \quad \frac{\partial \mathcal{L}}{\partial Z_{ai}} = 0, \quad \frac{\partial \mathcal{L}}{\partial W_{pq}} = 0 \quad (3.5)$$

所以

$$\frac{d\omega}{d\xi} = \frac{\partial \mathcal{L}}{\partial \xi} + \frac{\partial \mathcal{L}}{\partial X_{ia}^\sigma} \frac{\partial X_{ia}^\sigma}{\partial \xi} + \frac{\partial \mathcal{L}}{\partial C_{\mu p}^\sigma} \frac{\partial C_{\mu p}^\sigma}{\partial \xi} + \frac{\partial \mathcal{L}}{\partial \Omega} \frac{\partial \Omega}{\partial \xi} + \frac{\partial \mathcal{L}}{\partial Z_{ai}^\sigma} \frac{\partial Z_{ai}^\sigma}{\partial \xi} + \frac{\partial \mathcal{L}}{\partial W_{pq}^\sigma} \frac{\partial W_{pq}^\sigma}{\partial \xi} = \frac{d\mathcal{L}}{d\xi} \quad (3.6)$$

同样得到拉格朗日量对坐标的偏导数，由于仅考虑 TDA 所以其中一些项无需写为对称的形式。

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \xi} = & \sum_{\mu\nu} \left[\sum_{\sigma} (T_{\mu\nu}^\sigma + Z_{\mu\nu}^S) \frac{\partial h_{\mu\nu}^{core}}{\partial \xi} + \sum_{\sigma} (T_{\mu\nu}^\sigma + Z_{\mu\nu}^S) \frac{\partial g_{\mu\nu}^\sigma[\mathbf{D}]}{\partial \xi} + \sum_{\sigma} (T_{\mu\nu}^\sigma + Z_{\mu\nu}^S) v_{\mu\nu}^{xc,\sigma,(\xi)}[\rho] \right. \\ & \left. - c_x \sum_{\mu\nu} X_{\mu\nu}^{\beta\alpha} \frac{\partial K_{\mu\nu}^{\beta\alpha}[\mathbf{X}]}{\partial \xi} - \sum_{\sigma} \frac{\partial S_{\mu\nu}}{\partial \xi} W_{\mu\nu} \right] \end{aligned} \quad (3.7)$$

其中

$$\begin{aligned} T_{\mu\nu}^\sigma &= \sum_{ab} C_{\mu a}^\alpha T_{ab} C_{\nu b}^\alpha + \sum_{ij} C_{\mu i}^\beta T_{ij} C_{\nu}^\beta \\ T_{ab} &= \sum_i X_{ia} X_{ib} \\ T_{ij} &= - \sum_a X_{ia} X_{ja} \\ Z_{\mu\nu}^S &= \frac{1}{2} (Z_{\mu\nu} + Z_{\nu\mu}) = Z_{\nu\mu}^S \\ Z_{\mu\nu} &= \sum_{ai} C_{\mu i} C_{\nu a} Z_{ai} + \sum_{at} C_{\mu t} C_{\nu a} Z_{ai} + \sum_{ti} C_{\mu t} C_{\nu i} Z_{ti} \\ g_{\mu\nu}^\sigma[\mathbf{V}] &= \sum_{\kappa\lambda\tau} [(\mu\nu|\kappa\lambda) - c_x \delta_{\sigma\tau} (\mu\lambda|\kappa\nu)] V_{\kappa\lambda} \\ K_{\mu\nu}^{\beta\alpha}[\mathbf{X}] &= \sum_{\kappa\lambda} (\mu\lambda|\kappa\nu) \sum_{jb} C_{\kappa j}^\beta C_{\lambda b}^\alpha X_{jb} \\ v_{\mu\nu}^{xc,\sigma,(\xi)}[\rho] &= \frac{\partial v_{\mu\nu}^{xc,\sigma}[\rho]}{\partial \xi} \\ f_{\mu\nu,\kappa\lambda}^{xc,\sigma\tau,(\xi)}[\rho] &= \frac{\partial f_{\mu\nu,\kappa\lambda}^{xc,\sigma\tau}[\rho]}{\partial \xi} \end{aligned} \quad (3.8)$$

同样定义

$$G_{\mu\nu}^\sigma[\mathbf{V}] = g_{\mu\nu}^\sigma[\mathbf{V}] + f_{\mu\nu}^{xc,\sigma}[\mathbf{V}] \quad (3.9)$$

同样计算 $\sum_{\mu'} \frac{\partial \mathcal{L}}{\partial C_{\mu'p'}^{\sigma'}} C_{\mu'q'}^{\sigma'} = 0$ 。首先计算 $Q_{pq}^{\sigma'}$

$$\begin{aligned} Q_{i'j'}^{\sigma'} &= 2 \left(\sum_j T_{i'j}^{\sigma'} F_{j'j}^{\sigma'} + G_{i'j'}[\mathbf{T}] + \sum_a X_{i'a}^{\sigma'} G_{j'a}^{\sigma'}[\mathbf{X}] + g_{i'j'}^{xc,\sigma'}[\mathbf{X}, \mathbf{X}] \right) \\ Q_{a'b'}^{\sigma'} &= 2 \left(\sum_b T_{a'b}^{\sigma'} F_{b'b}^{\sigma'} + \sum_i X_{ia'}^{\sigma'} G_{ib'}^{\sigma'}[\mathbf{X}] \right) \\ Q_{i'a'}^{\sigma'} &= 2 \left(\sum_j T_{i'j}^{\sigma'} F_{a'j}^{\sigma'} + G_{i'a'}[\mathbf{T}] + \sum_a X_{i'a}^{\sigma'} G_{a'a}^{\sigma'}[\mathbf{X}] + g_{i'a'}^{xc,\sigma'}[\mathbf{X}, \mathbf{X}] \right) \\ Q_{a'i'}^{\sigma'} &= 2 \left(\sum_b T_{a'b}^{\sigma'} F_{i'b}^{\sigma'} + \sum_i X_{ia'}^{\sigma'} G_{ii'}^{\sigma'}[\mathbf{X}] \right) \end{aligned} \quad (3.10)$$

然后得到

$$\begin{aligned} \sum_{\mu'} \frac{\partial \mathcal{L}}{\partial C_{\mu'i'}^{\sigma'}} C_{\mu'j'}^{\sigma'} &= Q_{i'j'} + 2G_{i'j'}[\mathbf{Z}^S] - 2W_{i'j'} \\ \sum_{\mu'} \frac{\partial \mathcal{L}}{\partial C_{\mu'a'}^{\sigma'}} C_{\mu'b'}^{\sigma'} &= Q_{a'b'} - 2W_{a'b'} \\ \sum_{\mu'} \frac{\partial \mathcal{L}}{\partial C_{\mu't'}^{\sigma'}} C_{\mu'u'}^{\sigma'} &= Q_{t'u'} + 2G_{t'u'}^{\alpha'}[\mathbf{Z}^S] - 2W_{t'u'} \\ \sum_{\mu'} \frac{\partial \mathcal{L}}{\partial C_{\mu'i'}^{\sigma'}} C_{\mu'a'}^{\sigma'} &= Q_{i'a'} + \sum_{a\sigma'} Z_{ai'} F_{aa'}^{\sigma'} + \sum_t Z_{ti'} F_{ta'}^{\beta'} + 2G_{i'a'}[\mathbf{Z}^S] - 2W_{i'a'} \\ \sum_{\mu'} \frac{\partial \mathcal{L}}{\partial C_{\mu'i'}^{\sigma'}} C_{\mu't'}^{\sigma'} &= Q_{i't'} + \sum_{a\sigma'} Z_{ai'} F_{at'}^{\sigma'} + \sum_t Z_{ti'} F_{tt'}^{\beta'} + 2G_{i't'}[\mathbf{Z}^S] - 2W_{i't'} \\ \sum_{\mu'} \frac{\partial \mathcal{L}}{\partial C_{\mu'a'}^{\sigma'}} C_{\mu'i'}^{\sigma'} &= Q_{a'i'} + \sum_{i\sigma'} Z_{a'i} F_{i'i}^{\sigma'} + \sum_t Z_{a't} F_{i't}^{\alpha'} - 2W_{a'i'} \\ \sum_{\mu'} \frac{\partial \mathcal{L}}{\partial C_{\mu'a'}^{\sigma'}} C_{\mu't'}^{\sigma'} &= Q_{a't'} + \sum_{i\sigma'} Z_{a'i} F_{t'i}^{\sigma'} + \sum_t Z_{a't} F_{t't}^{\alpha'} - 2W_{a't'} \\ \sum_{\mu'} \frac{\partial \mathcal{L}}{\partial C_{\mu't'}^{\sigma'}} C_{\mu'i'}^{\sigma'} &= Q_{t'i'} + \sum_a Z_{at'} F_{ai'}^{\alpha'} + \sum_i Z_{t'i} F_{i'i}^{\beta'} + 2G_{t'i'}^{\alpha'}[\mathbf{Z}^S] - 2W_{t'i'} \\ \sum_{\mu'} \frac{\partial \mathcal{L}}{\partial C_{\mu't'}^{\sigma'}} C_{\mu'a'}^{\sigma'} &= Q_{t'a'} + \sum_a Z_{at'} F_{aa'}^{\alpha'} + \sum_i Z_{t'i} F_{a'i}^{\beta'} + 2G_{t'a'}^{\alpha'}[\mathbf{Z}^S] - 2W_{t'a'} \end{aligned} \quad (3.11)$$

对于 ROKS 有 $W_{pq} = W_{qp}$ ，同样可以得到 Z vector equation

$$\begin{aligned} -(Q_{i'a'} - Q_{a'i'}) &= \sum_{i\sigma'} Z_{a'i} F_{i'i}^{\sigma'} + \sum_t Z_{a't} F_{i't}^{\alpha'} - \sum_{a\sigma'} Z_{ai'} F_{aa'}^{\sigma'} - \sum_t Z_{ti'} F_{ta'}^{\beta'} - 2G_{i'a'}[\mathbf{Z}^S] \\ -(Q_{i't'} - Q_{t'i'}) &= \sum_a Z_{at'} F_{ai'}^{\alpha'} + \sum_i Z_{t'i} F_{i'i}^{\beta'} + 2G_{t'i'}^{\alpha'}[\mathbf{Z}^S] - \sum_{a\sigma'} Z_{ai'} F_{at'}^{\sigma'} - \sum_t Z_{ti'} F_{tt'}^{\beta'} - 2G_{i't'}[\mathbf{Z}^S] \\ -(Q_{t'a'} - Q_{a't'}) &= \sum_{i\sigma'} Z_{a'i} F_{t'i}^{\sigma'} + \sum_t Z_{a't} F_{t't}^{\alpha'} - \sum_a Z_{at'} F_{aa'}^{\alpha'} - \sum_i Z_{t'i} F_{a'i}^{\beta'} - 2G_{t'a'}^{\alpha'}[\mathbf{Z}^S] \end{aligned} \quad (3.12)$$

求解 Z vector equation 可以得到 Z_{ai}, Z_{at}, Z_{ti} , 进而可以得到 W_{pq}

$$\begin{aligned}
W_{i'j'} &= \frac{1}{2}Q_{i'j'} + G_{i'j'}[\mathbf{Z}^S] \\
W_{a'b'} &= \frac{1}{2}Q_{a'b'} \\
W_{t'u'} &= \frac{1}{2}Q_{t'u'} + G_{t'u'}^{\alpha'}[\mathbf{Z}^S] \\
W_{i'a'} &= \frac{1}{2}Q_{i'a'} + \frac{1}{2}\sum_a Z_{ai'}F_{a'a}^{\sigma'} + \frac{1}{2}\sum_t Z_{ti'}F_{ta'}^{\beta'} + G_{i'a'}[\mathbf{Z}^S] \\
W_{i't'} &= \frac{1}{2}Q_{i't'} + \frac{1}{2}\sum_{a\sigma'} Z_{ai'}F_{at'}^{\sigma'} + \frac{1}{2}\sum_t Z_{ti'}F_{tt'}^{\beta'} + G_{i't'}[\mathbf{Z}^S] \\
W_{a'i'} &= \frac{1}{2}Q_{a'i'} + \frac{1}{2}\sum_i Z_{a'i}F_{i'i}^{\sigma'} + \frac{1}{2}\sum_t Z_{a't}F_{i't}^{\alpha'} \\
W_{a't'} &= \frac{1}{2}Q_{a't'} + \frac{1}{2}\sum_{i\sigma'} Z_{a'i}F_{t'i}^{\sigma'} + \frac{1}{2}\sum_t Z_{a't}F_{t't}^{\alpha'} \\
W_{t'i'} &= \frac{1}{2}Q_{t'i'} + \frac{1}{2}\sum_a Z_{at'}F_{ai'}^{\alpha'} + \frac{1}{2}\sum_i Z_{t'i}F_{i'i}^{\beta'} + G_{t'i'}^{\alpha'}[\mathbf{Z}^S] \\
W_{t'a'} &= \frac{1}{2}Q_{t'a'} + \frac{1}{2}\sum_a Z_{at'}F_{aa'}^{\alpha'} + \frac{1}{2}\sum_i Z_{t'i}F_{a'i}^{\beta'} + G_{t'a'}^{\alpha'}[\mathbf{Z}^S]
\end{aligned} \tag{3.13}$$

有了 W_{pq} 和 Z_{pq} 带入 (3.7) , 然后加基态能量的梯度即可完成计算。